

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

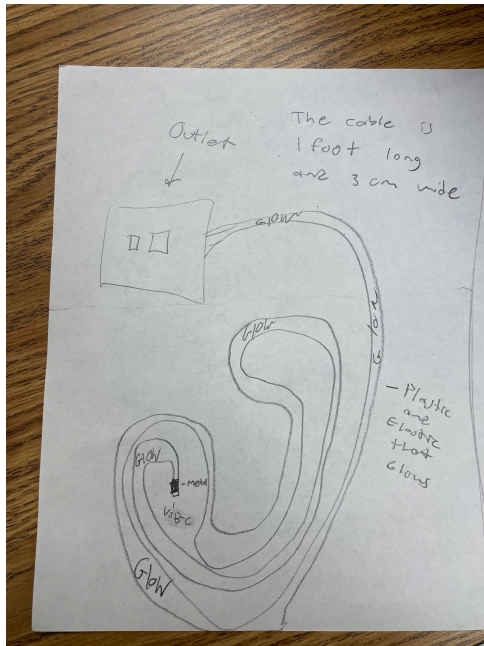
Design Idea Generation (D1)

Design Idea #1

Date: 03/06/2026

Description: Fluorescent phone charger - The charging cable glows in the dark and is a foot long to reach the charger straight down from the bed.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #2

Date: 03/06/2026

Description: Phone Holder - A device that holds the charger connection in place and the phone

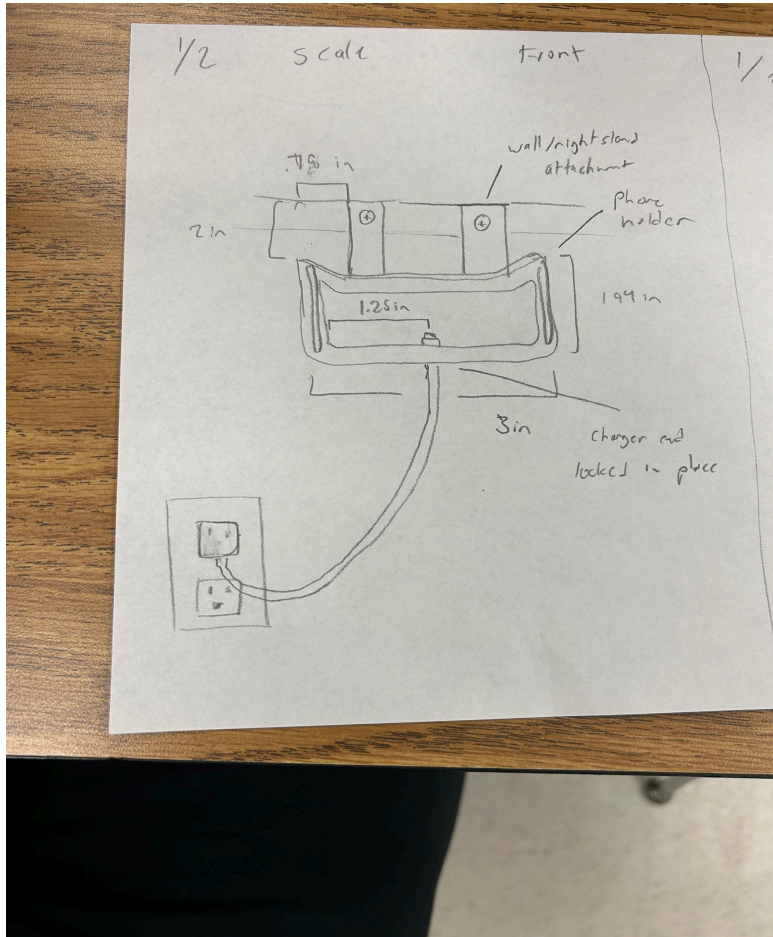
Name: _____

Date: _____

Class: _____

can be slotted into the charger. It is installed on a bedside or nightstand ledge and the phone can be inserted and aligned with the charging connection without adjustment. It also includes small light strips that can act as a nightlight if needed.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #3

Date: : 03/06/2026

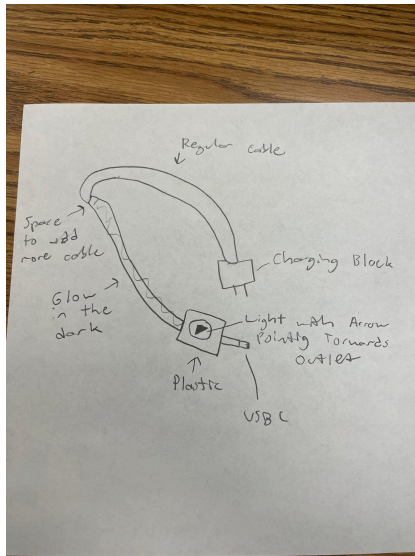
Description: A charging cable that is expandable depending if you need it to be longer or shorter. There is a screen on the cord that tells you which way is the outlet for when you need to plug your phone in the dark while in bed. The usb part also is magnetic with you phone for easy charging use.

Name: _____

Date: _____

Class: _____

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection (D1, D2)

Design Requirement	Fluorescent phone charger	Phone Holder	Charging Block Screen
Glows in the dark (1 pt)	Charging cable which glows in the dark	X	X
Design must be easily accessible in low visibility (3 pt)	Glow in the dark feature makes the cable visible	A device can be easily inserted into the holder and it has a nightlight feature	Charging block has an arrow with a lighted screen pointing towards the connection
Design is secured in one area (2 pts)	X	Devices can be placed in the slot and secured for	X

Name: _____

Date: _____

Class: _____

		the duration of charging	
Total Score	4 pts	5pts	3pt

Our team chooses design 2 to solve the community problem.

Design Blueprint (D4)

Design Blueprint #1

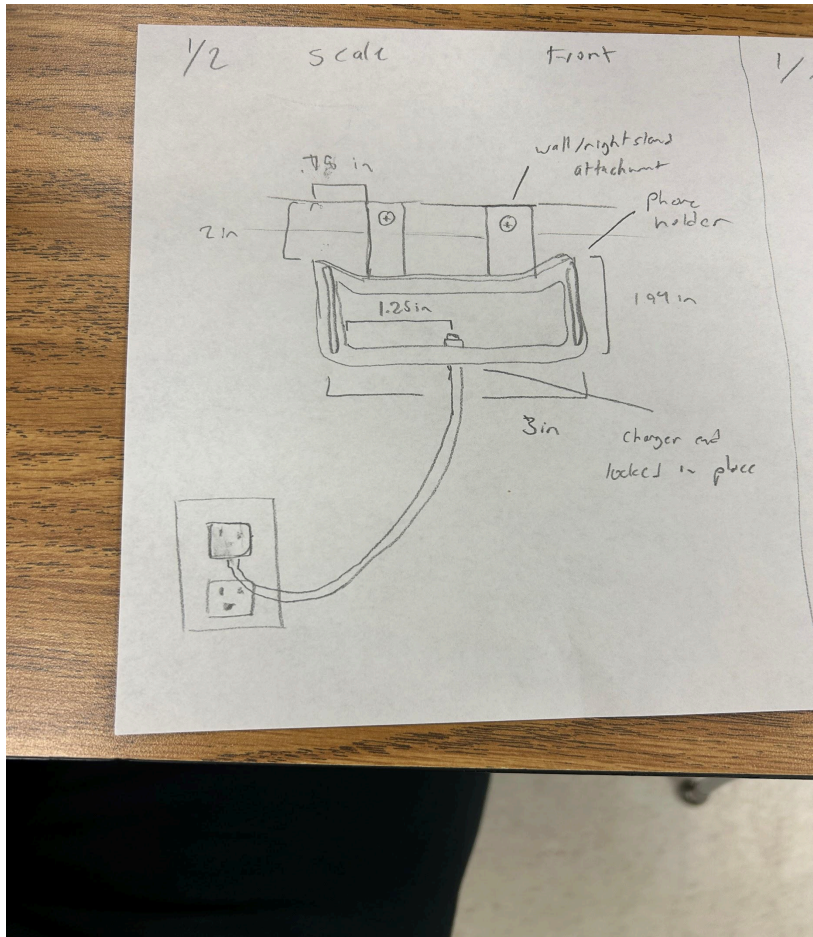
Date: 03/06/2026

Sketch:

Name: _____

Date: _____

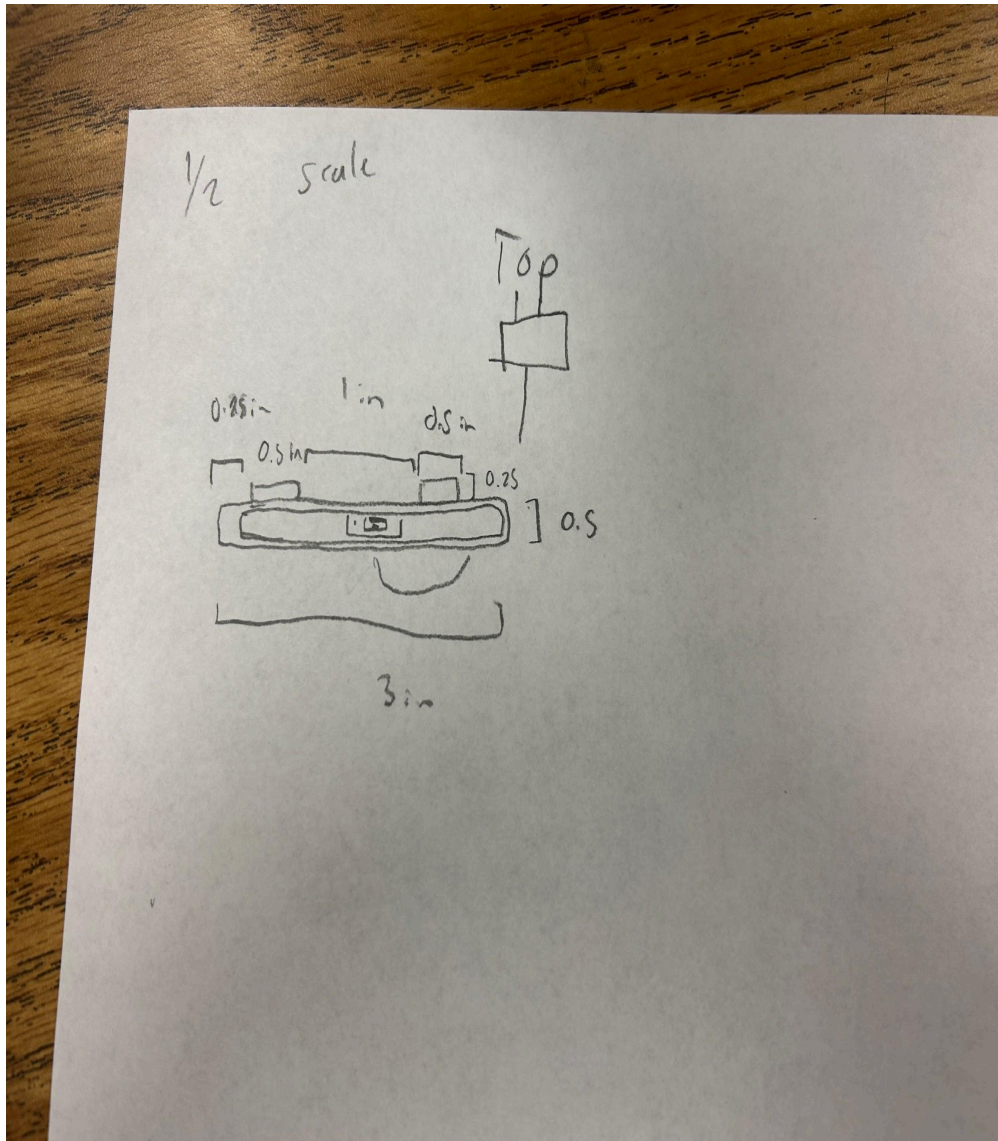
Class: _____



Name: _____

Date: _____

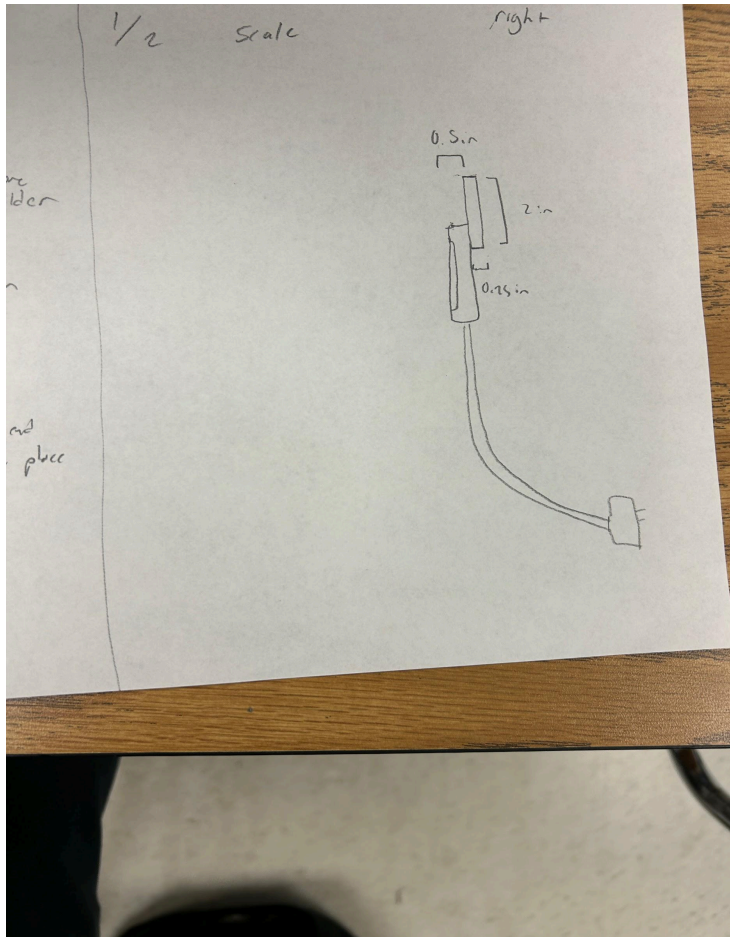
Class: _____



Name: _____

Date: _____

Class: _____



Manufacturing Methods:

Step 1: Sketch the base design in CAD and 3d print the base using plastic

Step 2: Add ambient light strips to the base

Step 3: Insert a charging cord into the proper point at the bottom of the

Supply List and Costs: (Not required for programming)

Supply List and Potential Costs:

Item	Cost each	# needed	Total
Ambient light strips	\$9	1	1
<1kg PLA filament (for 3d printing)	<\$13	1	1

Name: _____

Date: _____

Class: _____

Charging cable	\$5	1	1
Total Cost	~\$14-20		

Plan of Action (D3)

https://docs.google.com/document/d/1_oJel7-nPwHkC8SHTOMMzu3TO57rfCpSIDKII7Bloc8/edit?tab=t.0

Iterations for the Solution:

How have you changed your prototype? Record each iteration here. You may copy this blueprint for additional iterations.

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was <i>comprehensive, iterative, and consistently</i> defensible (viable design highly likely)	was <i>thorough, iterative, and</i> generally defensible (viable design likely)	was <i>adequate, generally</i> iterative and defensible (<i>viable design is possible</i>)	was <i>partial or overly general</i> and only somewhat iterative and/or defensible (raising issues with the design viability)	was <i>incomplete or minimally</i> iterative and/or defensible (<i>raising issues with the design viability</i>)	was not evident that there was an attempt



Name: _____

Date: _____

Class: _____

2.Design Solution Choice. The Design Solution chosen _____	<i>was well-justified and demonstrated attention to all design requirements</i>	<i>was justified and demonstrated attention to most if not all design requirements</i>	<i>was explained with reference to at least some design requirements</i>	<i>was not sufficiently explained with reference to design requirements</i>	<i>lacked support related to design requirements</i>	<i>was not attempted or evidence not provided</i>
3. Plan of Action. The plan of action _____	<i>has considerable merit and would easily support repetition and testing for effectiveness by others</i>	<i>would support repetition and testing for effectiveness by others</i>	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	<i>cannot be executed or does not exist</i>

